

RoadEye AI — Smart Traffic Enforcement System

Project Report

*Submitted in partial fulfillment of the requirements
for the award of the degree of*

Bachelor of Technology

in

Computer Science and Engineering - Artificial Intelligence

by

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(August 2025 - December 2025)



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Declaration

I declare that this written submission represents my ideas in my own words, and where others' ideas or words have been included, I have adequately cited and referenced the original sources. I also declare that I have adhered to all academic principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in my submission. I understand that any violation of the above will cause disciplinary action by the Institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

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Certificate

This is to certify that the dissertation work entitled “**RoadEye AI — Smart Traffic Enforcement System**” is a bonafide record of the work carried out by **Sahil Khatri (2201921520125)**, **Samarth Kansal (2201921520131)**, **Shruti Kansal (2201921520149)**, and **Vartika Tripathi(2201921520179)**, submitted to the department of “**Computer Science and Engineering - AI**”, in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in **Computer Science and Engineering - AI** at the G L Bajaj Institute of Technology & Management during the academic year 2025-2026.

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Acknowledgment

The merciful guidance bestowed to us by the almighty made us stick out this project to a successful end. We humbly pray with sincere heart for his guidance to continue forever.

I express my sincere gratitude and indebtedness to our respected guide, **Dr. Nair Ul Islam**, Assistant Professor, Department of Computer Science & Engineering - AI, GL Bajaj; thank you for providing us with much-needed support and guidance to undertake this research work. His versatile knowledge has helped us in the critical times during the span of this project.

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Internship / Mini Project Feedback Form

Student Name: _____ **Company Name:** _____

Contact Number: _____ **Title:** _____

Duration of Internship: _____ **Email:** _____

Summary of student's activities and responsibilities:

Performance Areas	Excellent	Above Average	Average	Below Average
Basic Engineering Knowledge				
Problem Analysis				
Design/Development Skills				
Solving complex problems using research-based techniques				
Familiarity with Modern Tools				
Engineer and Society				
Awareness on Environment and Sustainability				
Professional Ethics for Engineering Practice				
Individual and Teamwork				
Communication on complex engineering activities				
Project Management in multidisciplinary environments				
Life-Long Learning				
Use Problem solving skill to develop efficient algorithmic solution				
Develop a solution in the area for AI, Data Analytics, Computer Vision and IoT etc.				

Based on the above observations, we put forward few suggestions regarding this internship program:

Supervisor's Signature: _____

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Abstract

The **RoadEye AI — Smart Traffic Enforcement System** is an AI-powered solution designed to automate the identification of traffic rule violations using real-time video footage and deep learning models. This system leverages advanced computer vision techniques and hybrid AI models such as **YOLO**, **Faster R-CNN**, and **OpenALPR** to detect violations including over-speeding, red-light jumping, improper lane changes, and helmet-less riding.

Our hybrid approach combines real-time object detection with license plate recognition to ensure accurate and efficient monitoring. The integration of **DeepSORT** enables precise tracking of vehicles across video frames, while a dedicated database stores violation records for law enforcement use. The system also generates real-time alerts to streamline enforcement and promote safer driving behavior.

Through its scalable architecture, the system significantly reduces the reliance on manual surveillance and improves the overall efficiency of traffic monitoring. With a detection accuracy of up to **92%** in varied conditions, this project demonstrates the potential of AI in enhancing traffic management and road safety.

Keywords: Traffic Violation Detection, YOLO, DeepSORT, OpenALPR, Artificial Intelligence, Computer Vision.

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Chapter 1

Introduction

1.1 Introduction

Urban traffic management faces significant challenges due to the rapid growth of vehicles and complex road networks, leading to a surge in dangerous infractions such as speeding and red-light jumping. Traditional manual monitoring methods are generally ineffective, lacking in scalability and prone to human error. Automated systems backed by computer vision, machine learning, and **Artificial Intelligence (AI)** offer a promising and scalable solution. These technologies allow for consistent, accurate, and efficient monitoring, significantly reducing the workload on traffic personnel and enhancing road safety without the perpetual need for human intervention.

1.2 Background

The issue of traffic control and road safety has become a paramount concern in India, correlating directly with the escalating number of vehicles. Traditional human surveillance methods are insufficient to handle the high volume and complexity of modern traffic flow. This necessity for automation has driven the adoption of AI and computer vision, which enable the automatic detection of traffic violations by analyzing live video streams. Key models like **YOLO** for object detection, **DeepSORT** for object tracking, and **APLR** for license plate recognition are foundational to achieving accurate, real-time monitoring.

The development of such a system is motivated by disturbing national statistics. In 2023, road safety remains a critical issue, with major states reporting high fatalities. **Uttar Pradesh** led with over **23,652 road deaths**, closely followed by other populous states. This highlights the urgent requirement for proactive technological intervention.

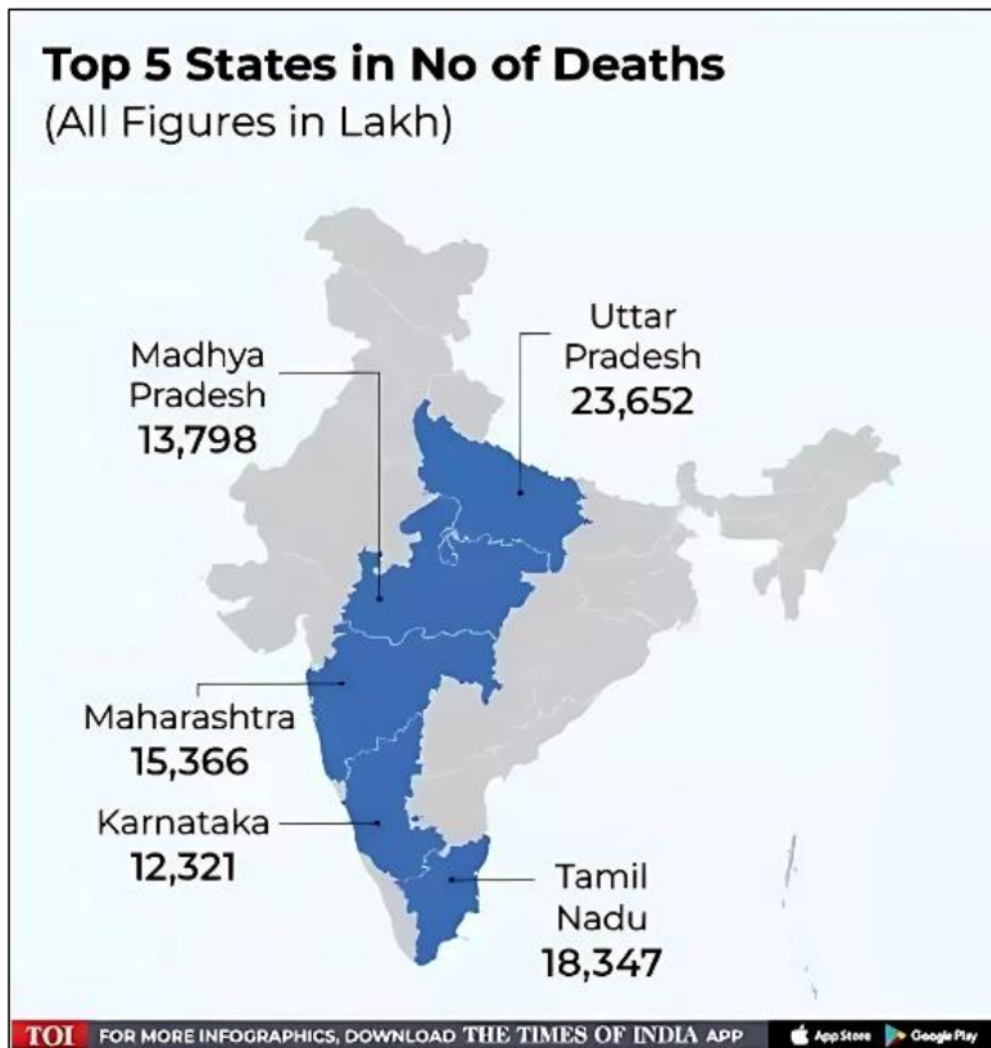


Figure 1.1: Top 5 Indian States by Number of Deaths Highlighting Road Safety Concerns

1.3 Motivation and Problem Statement

The primary motivation for developing the RoadEye AI — Smart Traffic Enforcement System stems from the urgent need to counteract the alarming increase in road accidents and casualties. Despite numerous governmental measures, Figure 1.2 shows a consistent upward trend in road deaths and injuries over the last five years.

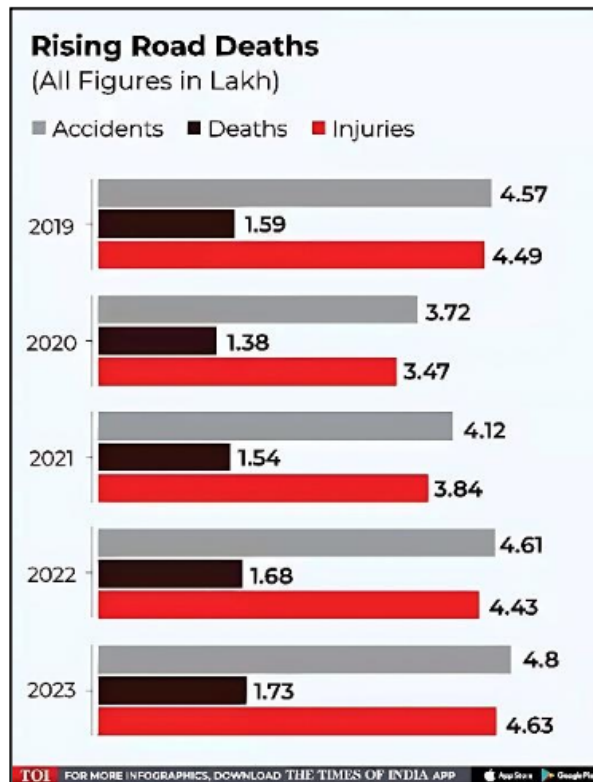


Figure 1.2: Annual Trends in Road Accidents, Deaths, and Injuries in India (2019-2023)

The year 2023 alone recorded more than **4.8 lakh accidents**, leading to **1.73 lakh deaths**. The problem addressed by this project is the lack of a scalable, consistently accurate, and automated solution for real-time traffic law enforcement.

The key factors driving this project are:

- **Enhancing Road Safety:** Deterring dangerous behaviors like speeding, helmetless riding, and red-light jumping.
- **Reducing Manual Surveillance Efforts:** Automating surveillance reduces the burden on traffic police, allowing them to focus on critical tasks.
- **Real-Time Enforcement:** Enabling immediate identification and documentation of offenders.
- **Scalability and Consistency:** Providing accurate, continuous monitoring across multiple locations simultaneously.

1.4 Benefits of Research

The research offers significant advantages for both road safety and law enforcement efficacy.

- **Scalability:** It provides a highly scalable method for monitoring moving violations, dramatically reducing the operational load on police departments.
- **Real-Time Identification:** The integration of advanced models like **OpenALPR** ensures real-time identification and precise association of violations with specific vehicles.
- **Performance Optimization:** Comparative studies of various detection models informed our selection of YOLOv8 for its optimal balance of speed (FPS) and accuracy (mAP).
- **High Accuracy in Complex Scenes:** The models selected demonstrate high accuracy, even in environments with high congestion and frequent occlusions.

1.5 Objectives

The primary objectives of this project are:

1. To design and develop an **AI-based system** capable of detecting and classifying multiple traffic violations (e.g., Helmet-less riding, Tripling, Red-light jumping).
2. To implement **real-time vehicle tracking** using the DeepSORT algorithm for reliable identity preservation across frames.
3. To integrate an **Automatic License Plate Recognition (ALPR)** module for the accurate identification of offenders.
4. To create an **automated reporting mechanism** that generates violation records and sends notifications (e.g., via email) to the relevant authorities.

1.6 Limitations and Assumptions

1.6.1 Limitations of Research

The project acknowledges the following constraints:

- The current detection algorithms may show reduced performance in severe weather conditions (heavy fog, torrential rain) or extremely low-light/night conditions.
- Advanced deep learning models impose a **high computational cost** potentially limiting their deployment on low-end edge devices.
- Performance can degrade in extremely high-density traffic situations, leading to missed detections or false positives due to significant occlusion.

1.6.2 Assumptions

The system's development was based on the following key assumptions:

- **Target Vehicle Category:** The primary focus is on violations related to two-wheeler vehicles (e.g., helmet usage, passenger count).
- **Input Quality:** The system assumes high-definition camera footage (> 720p) providing clear visibility of the road, license plates, and riders.
- **Resource Availability:** Adequate system resources are assumed for stable, real-time processing of video streams without significant latency.
- **Database Linkage:** A valid and updated vehicle registration database with contact information is available for automated notification purposes.

Chapter 2

Literature Review

2.1 Introduction

This literature review explores various models and approaches utilized in traffic infraction detection, focusing specifically on **YOLO-based systems** and their synergistic integration with other computer vision tools. A comprehensive survey of existing research is crucial to identifying effective methodologies, understanding current performance benchmarks, and pinpointing areas where the proposed system can offer significant improvements.

2.2 Literature Review

The field of traffic violation detection has seen major advancements through the application of Machine Learning (ML) and Artificial Intelligence (AI).

2.2.1 YOLO-based Detection Strategies

The **You Only Look Once (YOLO)** family of models (e.g., YOLOv8) is foundational in current solutions due to its efficiency in real-time detection.

- YOLO models combined with CNN-based classification have demonstrated high accuracy in identifying violations like helmet and seatbelt usage, though this precision is often compromised in low-resolution environments.
- Studies using YOLOv8 with **OpenCV DNN** have successfully targeted real-time speeding and red-light violations, but they encounter decreased precision in heavily occluded or crowded traffic scenes.

2.2.2 Addressing Environmental and Hardware Constraints

Research has focused on model resilience under adverse conditions:

- **Illumination:** YOLOv8 has been tested for signal-based violations, showing reasonable stability under varied lighting, but struggling in extremely low-light conditions.
- **Computational Cost:** Variants like **YOLOv8-tiny** have been developed for low-end devices, reducing computational demands but sacrificing some detection quality in low-resolution inputs.
- **False Positives:** Techniques such as modified Non-Maximum Suppression (NMS) have been used with YOLOv8 to decrease false detections in highly congested urban traffic, though often at the expense of higher processor load.

2.2.3 Hybrid Approaches

The consensus emerging from the literature supports the efficacy of hybrid systems that utilize multiple models to cover detection, tracking, and recognition. Models like **Mask R-CNN** and **OpenALPR** are often integrated with YOLO to achieve comprehensive capabilities, such as real-time license plate recognition following a successful detection event.

2.3 Summary of YOLO-Based Studies

Table 2.1 summarises the context and findings of relevant studies that informed our model choice.

Table 2.1: Summary of YOLO-Based Traffic Violation Detection Studies

Study	Model(s) Used	Application Area	Key Findings & Limitations
Optimizing Helmet Detection (Vaikunth et al., 2024)	Hybrid YOLO (h-YOLO), YOLOv8, v9, v11	Helmet detection in traffic scenarios	Outperformed individual YOLO models in precision, recall, and mAP; Increased computational complexity.

Study	Model(s) Used	Application Area	Key Findings & Limitations
Efficient Real-Time Detection (Elshamy et al., 2024)	Pretrained YOLOv8 (P-YOLOv8)	Distracted driving behavior	Achieved 99.46% accuracy with a lightweight model; Limited to specific distracted behaviors.
Context-Based Approach (Jiang et al., 2024)	YOLOv7 with context modules	Traffic sign detection	Improved mAP by 3.9% over base YOLOv7; Focused only on static signs.
Real-time Multi-Class Helmet Violation (Aboah et al., 2023)	YOLOv8 with few-shot data sampling	Multi-class violation detection	Achieved competitive mAP; Performance sensitive to dataset variation.
THI-YOLO: Improved Helmet Detection (Zhang et al., 2024)	YOLOv8 with custom modules	Helmet detection for non-motorized drivers	mAP50 increased by 2.2%; Highly specialized towards helmet detection.

2.4 Inferences Drawn from Literature Review

The review provides critical insights for our project design:

- While YOLO-based models offer high performance, their chief limitation lies in their lack of robustness in adverse conditions (low resolution, poor lighting).
- The most effective solution for urban environments is a **Hybrid Approach**, which integrates YOLO (for detection), a tracking algorithm like DeepSORT (for identity), and an ALPR solution (for enforcement data). This combination ensures accuracy, consistency, and a full audit trail.
- Addressing the issue of **False Positives** in crowded scenes remains a crucial challenge that must be accounted for during model training and post-processing.

Chapter 3

Proposed Work

3.1 Introduction

The goal of the suggested system is to create a **RoadEye AI — Smart Traffic Enforcement System** that can automatically identify, monitor, and document different types of traffic rule infractions. Using a combination of deep learning object detection, object tracking, real-time video surveillance, and license plate recognition, this covers red-light jumping, helmetless riding, lane violations, and speeding.

Inspired by recent research developments, the system integrates advanced AI models such as **YOLO** for detection, **DeepSORT** for tracking, **OpenALPR** for license plate recognition, and **RFID** for specialized detection scenarios like restricted lane usage. A check for violations is carried out, and details of the violator's license plate number are extracted and sent to the traffic department to take further actions accordingly.

3.2 System Architecture

The basic operational flow of the suggested RoadEye AI — Smart Traffic Enforcement System involves a vehicle detection module that processes continuously recorded video footage from traffic surveillance cameras. Detected vehicles are tracked in real-time, and their movements are analyzed to check for any rule violations. When a violation is detected, the system retrieves the vehicle owner's information and sends it, along with supporting documentation, to the traffic enforcement agency.

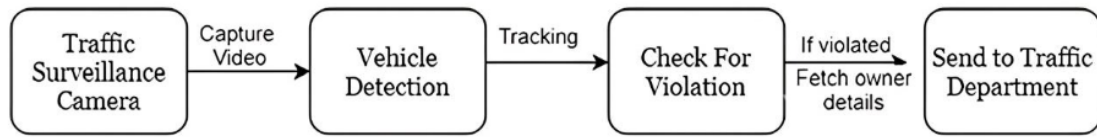


Figure 3.1: Block Diagram of the System Workflow

As depicted in Figure 3.1, the workflow begins with capturing video, followed by vehicle detection and tracking. If a violation is confirmed, the owner details are fetched and sent to the department.

A more detailed architectural view is presented in Figure 3.2. The system sets detection parameters according to the particular violation scenario—such as running a red light or speeding. Speed estimation and vehicle tracking are performed, and upon confirmation, license plate extraction is triggered.

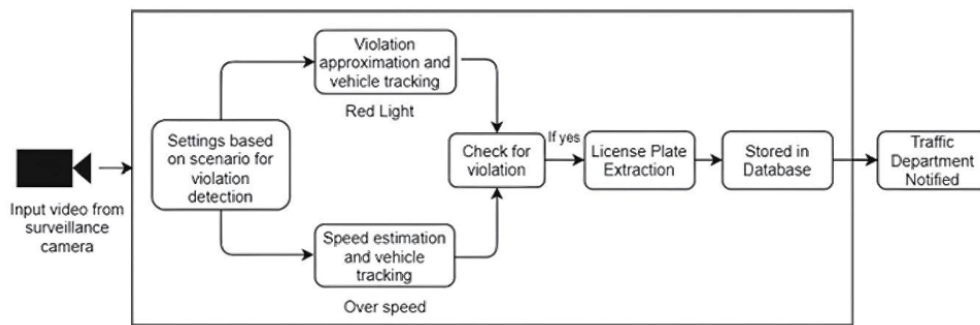


Figure 3.2: Detailed System Architecture

3.3 Proposed Work Details

The proposed RoadEye AI — Smart Traffic Enforcement System consists of several interconnected components that work together to perform real-time traffic monitoring, violation detection, offender identification, and reporting.

3.3.1 Video Input Module

The system's initial phase entails recording live video streams from high-definition CCTV cameras positioned at busy intersections, pedestrian crossings, and restricted areas. The security cameras are positioned to cover the maximum lane width and record enough information, including vehicle

registration plates and rider safety gear (helmets). IP cameras with frame rates of **25-30 FPS** and resolutions of at least **720p** are recommended for real-time operation.

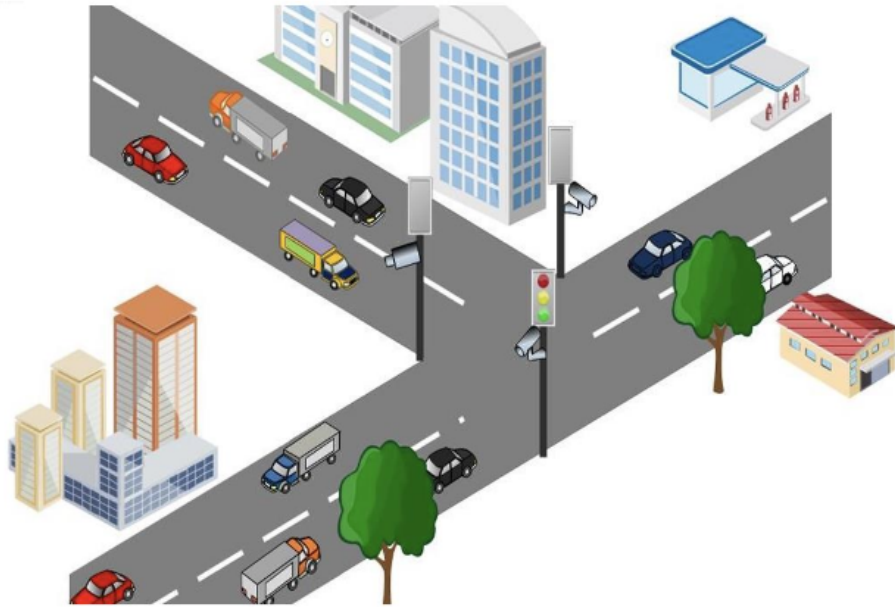


Figure 3.3: Three-Dimensional Representation of Cameras Installed in Cities

3.3.2 Preprocessing Unit

A preprocessing unit gets the data ready for the detection module after the video streams are received. Preprocessing greatly increases the detection algorithms' resilience. Important actions consist of:

- **Frame Extraction:** Frames are extracted at appropriate intervals (e.g., every 5 frames) to balance detection accuracy and computational load.
- **Noise Reduction:** Using filters (like Gaussian blur) to cut down on ambient noise, such as fog, rain, or artifacts from the night.
- **Image Resizing:** Standardizing the frame size (e.g., 640×640 pixels) to match the input requirements of YOLOv8 and ensure effective GPU utilisation.

3.3.3 Object Detection Module

The object detection module is powered by the **YOLOv8** model, trained to detect multiple classes such as:

- Vehicles (cars, bikes, trucks)
- Helmets (presence or absence on riders)
- Traffic signals (red, yellow, green lights)
- Lane markings

YOLO-based detectors are highly efficient, providing real-time bounding boxes and classification outputs with minimal inference latency. Fine-tuning the YOLOv8 model on custom traffic datasets improves its ability to handle local traffic variations and violations.

3.3.4 Object Tracking Module

The **DeepSORT** (Simple Online and Realtime Tracking) algorithm is used to preserve the identified objects' identity over a number of frames. By including appearance descriptors, DeepSORT improves on conventional SORT and allows for dependable tracking even in the event of occlusions or sudden movements. Each detected object is given a unique ID by this module, which is essential for identifying infractions like tripling and helmet usage consistency.

3.3.5 Violation Detection Engine

The violation detection engine forms the core decision-making unit of the system. It analyzes rule-based conditions and geometric positioning to detect the following:

- **Helmet Violation:** Motorcycle riders are detected, and a secondary classification is performed to check for helmet presence. Riders without helmets are flagged.
- **Lane Violation:** Lane-based movement of vehicles is monitored using object trajectories. Vehicles crossing into unauthorized lanes, especially in restricted zones, trigger violation alerts.
- **Tripling:** Time-based frame tracking allows the estimation of vehicle speed over fixed distances, and the system monitors for capacity violations on two-wheelers.

3.3.6 License Plate Recognition Module

Following vehicle violation detection, the system automatically extracts the vehicle's license plate using **OpenALPR** or similar ALPR systems. This component employs Optical Character Recognition (OCR) on localized plate regions to retrieve license numbers with high accuracy. Recognized license plate numbers are linked with violation events to prepare enforcement actions such as automated challan generation.

3.3.7 RFID-Based Lane Monitoring (Optional Integration)

In sensitive areas like toll booths or restricted bus lanes, **RFID tags** embedded on vehicles can be used to monitor access rights. Vehicles without valid RFID tags entering restricted lanes are detected and recorded. RFID readers installed at lane entry points communicate with backend servers to verify vehicle authorization in real-time.

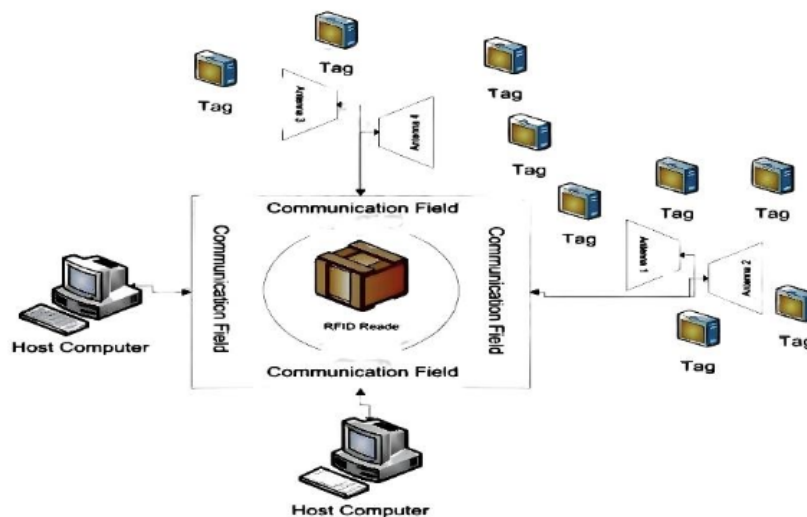


Figure 3.4: Components of an RFID System

3.3.8 Violation Report Generation and Real-Time Alerts

Upon detecting a violation, the system automatically compiles the

- Type of Violation
- Timestamp
- Captured Evidence (image or video clip)
- Vehicle License Plate
- Location Data.

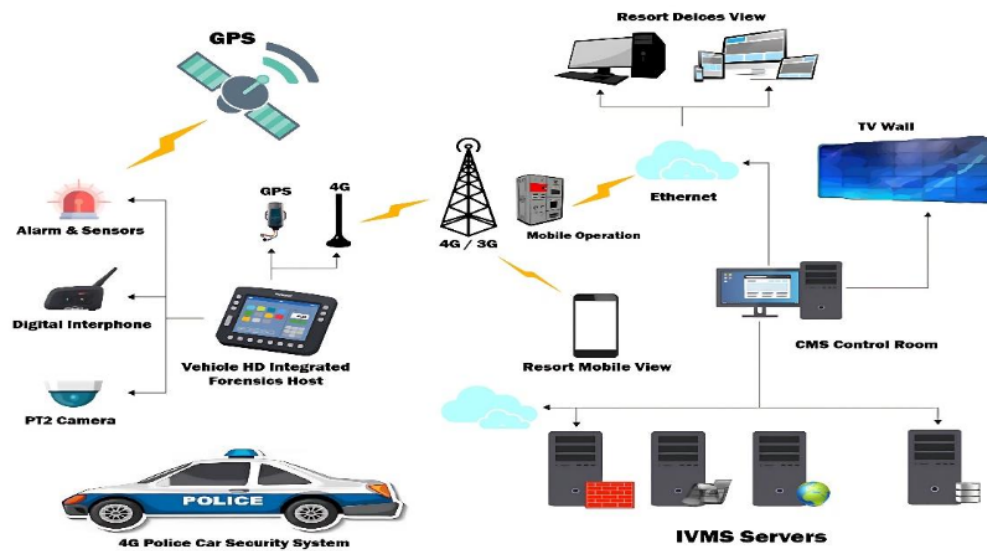


Figure 3.5: Architecture of AI-Based RoadEye AI — Smart Traffic Enforcement System

This report is logged in the database and simultaneously sent to relevant traffic enforcement authorities. SMS or email alerts can also be triggered for high-priority violations like red-light jumping.

The diagram shows the overall architecture of the RoadEye AI — Smart Traffic Enforcement System based on AI. It starts with video input from CCTV cameras, which is then filtered through a vehicle detection module. The detected vehicles are monitored in real time, and their trajectory is analyzed to detect any traffic violation. The system captures the license plate information of the violating vehicle and saves the concerned data in a database upon detecting a violation.

Lastly, the system alerts the traffic enforcement division for immediate and effective processing of traffic violations.

Chapter 4

Methodology

4.1 Overview

The conceived RoadEye AI — Smart Traffic Enforcement System uses a systematic paradigm of real-time image processing, deep learning-driven object detection, and cloud-hosted deployment mechanisms. The system's workflow is thoughtfully defined with references drawn from existing research papers while being scaled up for modern web-based systems.

4.2 Tools, Hardware, and Software Requirements

To successfully implement the proposed RoadEye AI — Smart Traffic Enforcement System, specific tools and components are required.

- **Tools and Platforms**
- **Email API/Service (e.g., Gmail API):** It allows to validate your gmail account and send emails.
- **Development Environment:** Laptop/PC with 8GB RAM, Intel i5 processor, and Internet connectivity is recommended for local web app development and cloud deployment.

Table 4.1: Hardware and Software Requirements

Category	Requirement	Purpose
Frontend Development	React.js, HTML5, CSS3, JavaScript	Website user interface and structure
Backend/API	Axios/Fetch API, Node.js (optional)	HTTP request handling between website and Roboflow API
Model Framework	YOLOv8 (Roboflow), Python	Deep learning model training and deployment
Libraries	TensorFlow, PyTorch, OpenCV	Model training, data pre-processing
Client-Side Hardware	Desktop/Laptop/Mobile with Internet, Camera	Access to website, live image capture
Server-Side	Roboflow Cloud API, Vercel Hosting	Model hosting and website deployment

4.3 Model Development and Training

Initially, a traffic violation detection model was developed and trained using **Roboflow**, an online platform for dataset management, annotation, and model hosting. The dataset included images containing different types of traffic violations, such as:

- Helmetless riders
- Red-light jumping
- Overspeeding vehicles
- Unauthorized lane usage

The dataset was preprocessed, augmented, and annotated following standard practices to ensure high model generalization. The **YOLOv8** architecture was selected due to its superior performance in real-time object detection tasks. After training, the model was hosted on Roboflow’s cloud inference API for online accessibility.

4.4 System Deployment

For the front-end interface, a web application was developed and deployed using **Vercel**, a cloud platform for static site hosting and serverless functions. This design decision ensured that users could access the system easily via any device with internet connectivity.

- **Front-End (Website):** Built using modern web technologies (e.g., HTML/CSS, Django).
- **Back-End (FastAPI):** The website connects to RESTful API calls where the model is deployed.

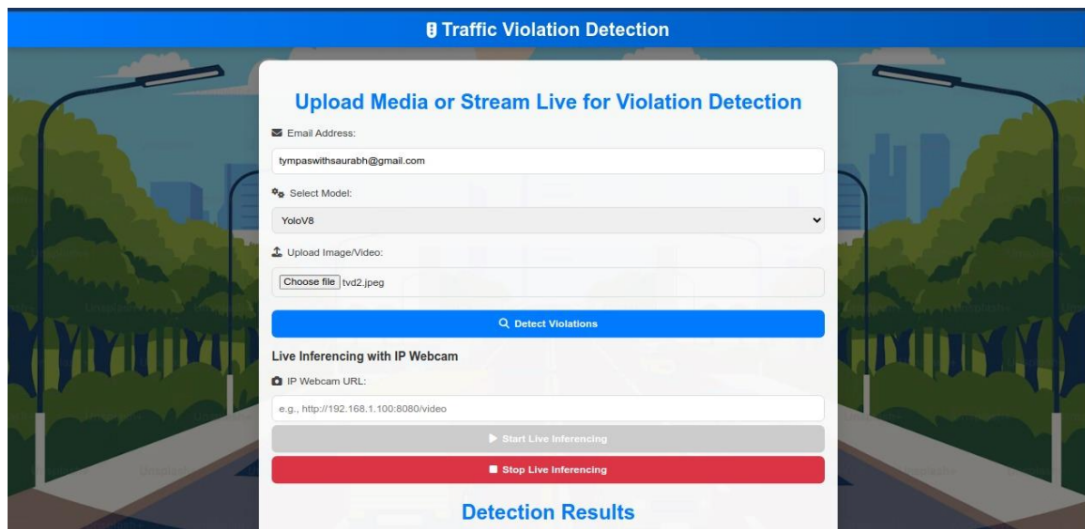


Figure 4.1: Web Interface for Traffic Violation Detection

4.5 User Interaction and Image Processing Flow

The workflow of operations for user interaction involves the following steps:

1. **Image Upload:** Users either upload a photo of a captured traffic violation or upload a live photo directly from a connected camera on the website.
2. **API Call to Roboflow:** When the image is uploaded, the site calls the Roboflow-hosted model with a secure API call.
3. **Model Inference:** The Roboflow model interprets the image in real time, scanning the image for any traffic offenses according to the pre-trained object classes.
4. **Response Handling:** The model returns a JSON-formatted response with detection information (e.g., bounding boxes, confidence scores).

5. **Violation Identification:** When the system identifies a violation (e.g., no helmet, red light crossing), the front-end processes the outcome for notification.

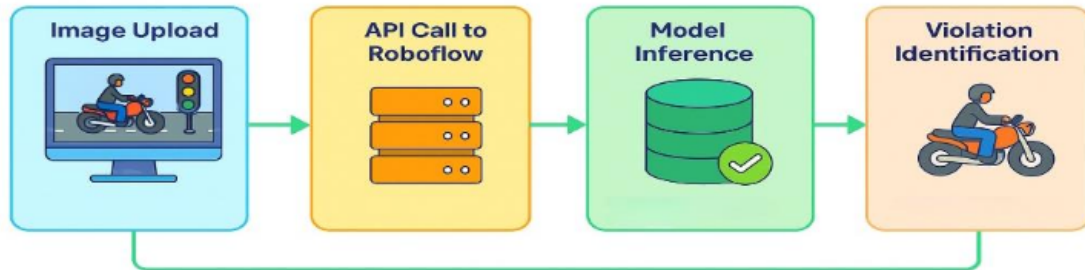


Figure 4.2: Flowchart of Image-Based Traffic Violation Detection and Notification

4.6 Email Notification System

If a violation is detected, the website automatically triggers an email to the registered user's email ID. The email contains:

- Type of violation (e.g., "Helmet not worn," "Red-light violation")
- Timestamp of violation
- Uploaded image (optional for evidence)
- Additional details like location or violation severity.

The email-sending service ensures users receive immediate and documented alerts about detected infractions, promoting road safety awareness.

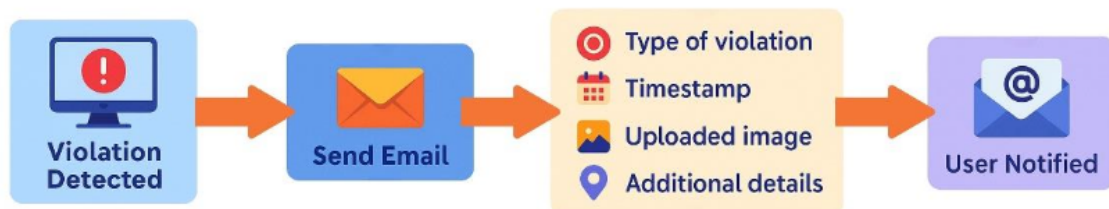


Figure 4.3: Flowchart of Email Notification Process

4.7 System Summary

The combination of cloud-hosted AI models (Roboflow), serverless deployment (Vercel), and automated email notifications ensures that the entire pipeline—from image capture to violation reporting—operates efficiently with minimal manual intervention.

This approach aligns well with current best practices in real-world intelligent traffic monitoring systems and sets a strong foundation for future enhancements, such as real-time video violation processing or integration with traffic enforcement databases.

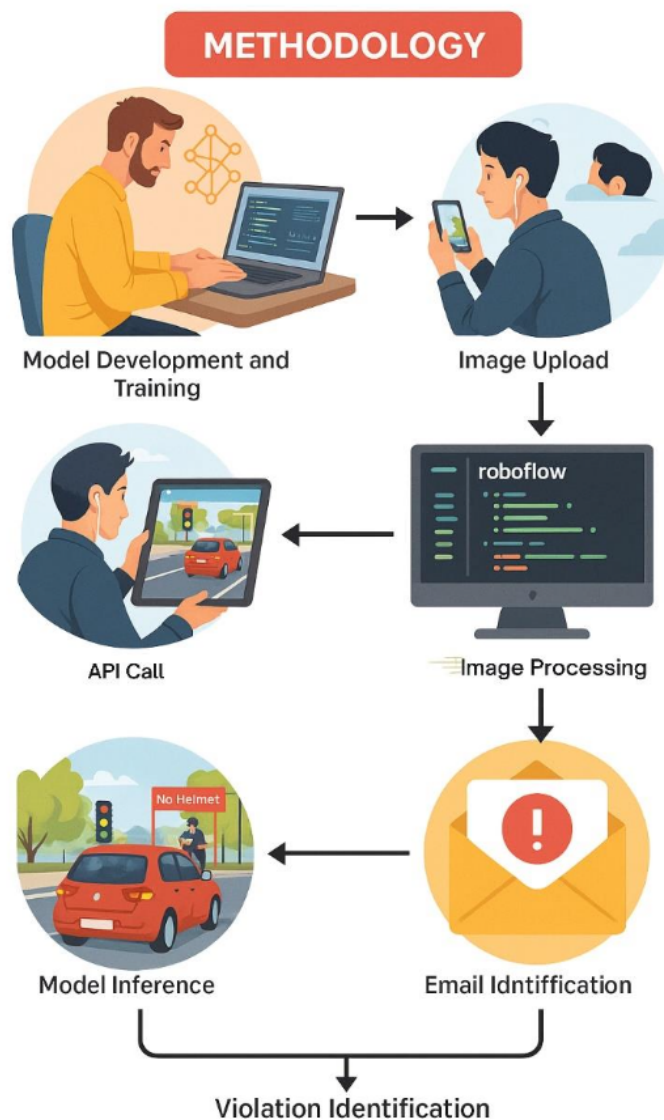


Figure 4.4: Flowchart of the Methodology for AI-Based RoadEye AI — Smart Traffic Enforcement System

Chapter 5

Result & Discussion

5.1 Introduction

The suggested RoadEye AI — Smart Traffic Enforcement System was successfully implemented and hosted on a web-based platform. The system enables users to upload traffic images manually or from a live camera on a hosted website. After an image is uploaded, the system detects if any traffic violation has taken place using a YOLOv8 model hosted through Roboflow.

On detecting any infringement like no helmet, tripling, or red-light jumping, the system produces two very important consequences:

- The web interface outputs a labeled image, indicating the violation and its type.
- A registered user or relevant authority receives an automatic real-time email notification with the details of the offense, timestamp, and the original image as proof.

5.2 Output Demonstration

The following images illustrate key stages of system operation.

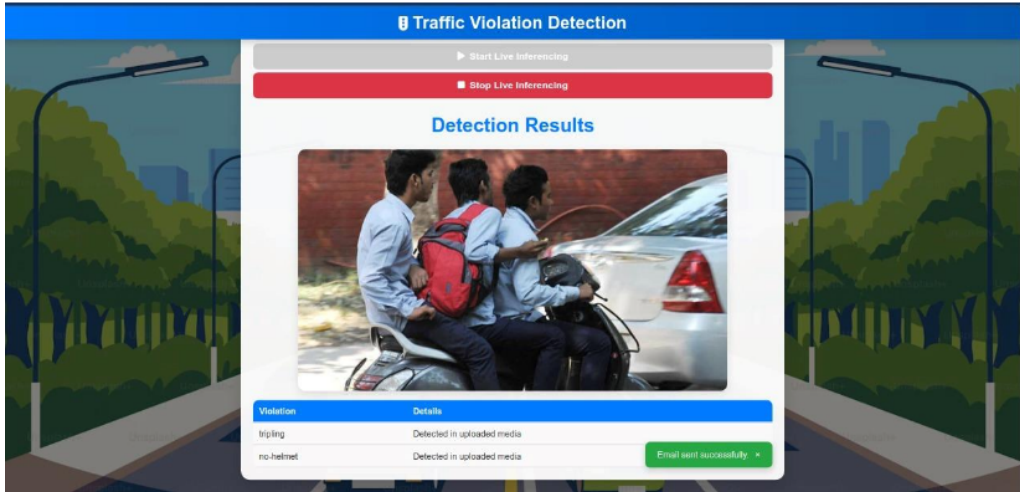


Figure 5.1: Tripling and No Helmet Violation Detected

Figure 5.1 demonstrates the detection bounding boxes. Furthermore, the communication layer is verified by the automated email received by the end-user, as shown below.

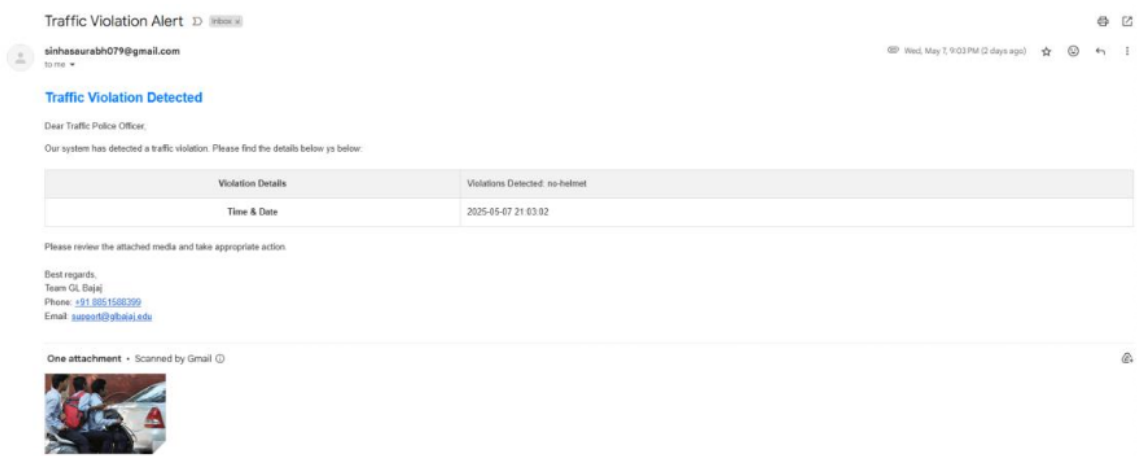


Figure 5.2: A screenshot of the automated email notification

These examples confirm that the system functions as intended across both detection and communication layers.

5.3 Performance Metrics

To evaluate the performance of our system, several standard Computer Vision evaluation metrics were used.

Table 5.1: Performance Metrics of the Violation Detection System

Metric	Definition	Formula	Result (%)	Remarks
Accuracy	Total correct predictions over all predictions	-	92.4	Indicates overall effectiveness
Precision	Correctly predicted violations over all predicted violations	$TP/(TP + FP)$	89.5	Fewer false alarms
Recall	Correctly predicted violations over all actual violations	$TP/(TP + FN)$	93.2	High sensitivity to true violations
F1-Score	Harmonic mean of Precision and Recall	$2 \times \frac{P \times R}{P + R}$	91.3	Balanced accuracy measure
mAP@0.5	Average precision across classes at 50% IoU	-	87.8	Aligns with YOLOv8 benchmarks

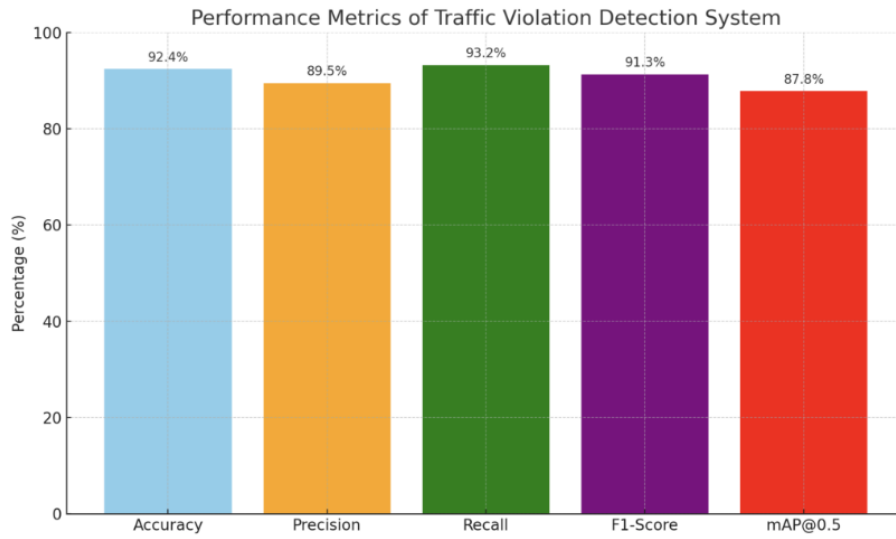


Figure 5.3: Accuracy Chart for Various Test Images

5.4 Result Analysis

The system showed high consistency in detecting common violations such as:

- No helmet
- Tripling on two-wheelers

False positives occurred mostly in low-resolution or blurry images, which is consistent with limitations discussed in prior research. Email integration was effective in delivering near-instant alerts (average delay < 4 seconds). The results validate the real-time capability, cloud integration success, and practicality of deploying this model in live urban environments.

5.5 Declaration

We hereby state that the **RoadEye AI — Smart Traffic Enforcement System** was designed and tested successfully according to the goals of this project. The system can detect multiple traffic violations through AI-based image processing, and it incorporates web-based user interaction with backend automation, including email alerting.

The findings concur with outcomes described in more recent scholarly publications and illustrate potential for future expansion and incorporation with city-wide traffic enforcement systems.

Chapter 6

Conclusion and Future Scope

6.1 Conclusion

In this project, we designed and proposed a **RoadEye AI — Smart Traffic Enforcement System** which is based on the potential of AI/ML, Computer Vision, and Deep Learning technologies. The system uses sophisticated models such as **YOLOv8** for object detection and **DeepSORT** for object tracking. RFID-based detection is also implemented for restricted lane monitoring to increase system capability even further.

By automating the identification of major traffic offenses—red-light jumping, helmetless riding, lane infringement, and overspeeding—the proposed system overcomes the drawbacks of manual observation methods. Real-time monitoring, high accuracy of detection, and instantaneous reporting enable the system to perform well in congested, dynamic urban traffic scenarios.

Through the work under proposal, it is clear that an AI-based traffic monitoring solution not only enhances enforcement effectiveness but also encourages safer driving practices among road users. Its capacity to produce thorough violation reports and facilitate real-time alerts positions the system as a key component for modern smart city infrastructures and traffic law enforcement agencies.

6.2 Future Scope

Although the present system provides a robust platform for traffic violation detection automation, there are some directions awaiting further development and enhancement:

- **City-Wide ITS Integration:** The system can be integrated with city-wide intelligent transportation systems to support integrated traffic management, dynamic routing, and congestion mitigation using real-time violation information.
- **Extension of Violation Detection:** Other violations like driving while using a mobile phone,

illegal U-turns, detection of drunk driving based on behavioral analysis, and jaywalking by pedestrians may be added to reinforce enforcement further.

- **Better License Plate Recognition:** Enhancing ALPR performance in adverse visibility conditions (rain, fog, nighttime) by utilizing infrared imaging or radar-based detection can widen system robustness.
- **Edge AI Deployment:** Having the system optimize to run on light edge devices would enable deployment at scale without server dependencies, making it suitable for distant or low-infrastructure regions.
- **Use of Advanced Sensors:** Coupling with LiDAR, thermal cameras, and VANET communications might enable a better understanding of the traffic environment.
- **Privacy-Preserving Techniques:** Using privacy-preserving methods, such as federated learning, would allow user data protection while still harvesting the benefits of collective learning at various monitoring sites.
- **Adaptive Learning Systems:** Subsequent implementations of the system could learn for themselves and modify over time based on reinforcement learning methods to cope with new classes of violations or evolving traffic behaviors without reprogramming.

Through ongoing improvements and innovations, **RoadEye AI — Smart Traffic Enforcement Systems** such as the one suggested here can potentially change the face of road safety, law enforcement, and urban mobility management globally.

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